

**Vaccination certificates in dogs and cats: immunological basis, health
validity and international regulatory framework**

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Abstract

Objective. To analyze the immunological basis of rabies vaccination certificates for dogs and cats, their international health validity, and the criteria determining their acceptance or rejection at border control. **Method.** Review of regulatory frameworks from major destination countries, primary and booster vaccination protocols, and identification and traceability requirements. **Results.** Certificate validity depends not solely on the vaccination date but on the concordance between microchip, species, approved vaccine product, complete primary schedule, and the time window required by the destination country. An error in any of these elements invalidates the document. **Conclusion.** Correct management of the vaccination certificate is the first critical link in the pet export file and determines the animal's admissibility regardless of its serological status.

Keywords: vaccination certificate, rabies, ISO microchip, primary schedule, traceability, pet export, document validity

Review of the immunological basis of animal vaccination, the relationship between biological and legal validity, common technical errors, and the comparative regulatory framework of the main pet export destinations.

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Section 1 The vaccination certificate as a medical-documentary act

There is a tendency, understandable but technically incorrect, to treat the vaccination certificate as an administrative procedure: a form to be filled in after administering a vaccine and whose function is to certify that fact before an authority. This understanding underestimates the nature of the document and, in the practice of international animal movement, is the root of many errors with consequences that are difficult to reverse.

The vaccination certificate is the documentary formalisation of a complex preventive medical act. That act has several components that must be fulfilled correctly and in the proper sequence for the document to have technical validity, not merely formal:

1.1 Unequivocal identification of the animal

The first component of the medical-documentary act is the identification of the individual to whom the vaccine is administered. In the context of international movement, the standard identification is the microchip conforming to ISO 11784/11785, whose number must appear on the certificate and match exactly with that read on the animal at any subsequent control point.

This concordance is not a bureaucratic requirement. It has its basis in a fundamental technical question: how can the health authority of the destination verify that the animal before it is the same one that received the registered vaccine? Without a permanent and unique identifier, that verification is not possible. The microchip number is the common thread that links the biological individual with the entire health documentary chain that accompanies it.

1.2 Correct administration of the vaccine

The veterinarian's signature on a vaccination certificate is a technical declaration, not a procedural rubber stamp. It implies that the professional confirms that: the vaccine administered corresponds to the species and indication, the vaccine was stored under appropriate cold chain conditions, the dose and route of administration were those indicated by the manufacturer, the animal was in clinical condition compatible with vaccination at the time of the act, and the batch number and expiry date have been correctly recorded when destination regulations so require.

Each of these points may be subject to verification by the destination health authorities. A certificate that cannot demonstrate that all were fulfilled has limited documentary robustness, regardless of its format.

1.3 Record compliant with current regulations

The same clinical act may produce a valid or invalid document depending solely on whether its record complies with the formal requirements of the country of destination. These requirements may include: the use of a specific official format (numbered, printed, with verification code), the transcription of the issuing veterinarian's authorisation or registration number, the signature and stamp of the official or accredited veterinarian when regulations so require, or registration in a national or regional registry system.

The clinical content may be impeccable and the document may still be rejected at the border if any of these formal requirements are not met. This apparent paradox has precise health logic: animal movement control systems are designed to be verifiable at scale, and that verifiability requires formal standardisation, not just clinical correctness.

Section 2 Immunological basis of animal vaccination

2.1 The vaccination act and protection: a necessary distinction

Vaccinating an animal is not synonymous with protecting it. It is initiating a biological process whose outcome --the generation of functional immunological memory-- depends on variables that the vaccination act alone does not control. Understanding this distinction is essential to correctly interpret both the immunological and legal validity of vaccination certificates for pet export.

The adaptive immune system, upon recognising the antigens contained in a vaccine, triggers a response that includes the proliferation and differentiation of specific B and T lymphocytes. Differentiated B lymphocytes produce antibodies of progressively more efficient classes (affinity maturation process), while a subpopulation of both lineages becomes long-lasting memory cells. It is these memory cells that confer real protection against subsequent exposure to the infectious agent.

2.2 Humoral response and immunological memory

The humoral response --mediated by antibodies-- is the immunological parameter most frequently measured in the context of vaccination for travel, because it is quantifiable by serological tests. After a first vaccination (primary response), specific antibody titres appear with a delay of days to weeks, reach a peak and then decline. This kinetic pattern is the immunological basis of the minimum post-vaccination waiting period required by import regulations: the organism needs time to generate a measurable response.

In a booster vaccination (secondary or anamnestic response), previously generated memory cells respond more quickly, intensely and durably. This difference has practical implications: an animal vaccinated for the first time does not have the same immediate protection as one with a prior vaccination history, although both may carry a formally valid vaccination certificate.

The WSAVA (World Small Animal Veterinary Association) vaccination guidelines explicitly distinguish between «core» vaccines --those whose administration is recommended to all animals of the species regardless of their situation-- and «non-core» vaccines, whose indication depends on individual epidemiological risk. The rabies vaccine

occupies a special position in the context of international movement that transcends this classification: it is a public health requirement with a legal basis in most destinations, regardless of the individual epidemiological risk of the animal.

2.3 Primary vaccine failure

Primary vaccine failure is the absence of an adequate immune response following a correctly performed vaccination. Its most frequent causes in clinical practice include:

- Maternal antibody interference: in puppies and kittens, antibodies transferred by the mother through colostrum may neutralise vaccine antigens before the offspring's immune system can recognise them and generate its own response. This is the basis for serial vaccination protocols in young animals.
- Concomitant immunosuppression: debilitating diseases, immunosuppressive treatments or active infections at the time of vaccination may compromise the resulting immune response.
- Cold chain deficiencies: many vaccines are heat-labile. Exposure to inadequate temperatures during transport or storage may degrade the antigens.
- Individual genetic variability: a small proportion of individuals of any species has reduced vaccine response capacity due to genetic factors that do not manifest clinically in any other way.

Primary vaccine failure is the reason why the antibody titration test --the RNATT-- exists as a regulatory requirement in certain destinations: it verifies that the immune response actually occurred, not merely that the vaccination act was performed.

Section 3 Rabies vaccination and public health

3.1 Epidemiological justification

Rabies is a viral encephalitis invariably fatal in unvaccinated mammals once clinical symptoms are established. Its causal agent, the rabies lyssavirus, has a host spectrum that

includes virtually all domestic and wild mammals. Transmission occurs mainly through the saliva of infected animals, by bite or contact with mucous membranes or open wounds.

The relevance of rabies in the international movement of companion animals does not derive from its prevalence in countries of origin --many are in countries with advanced control programmes or rabies-free status-- but from the consequences that introduction of the virus into currently free territories would have. Island countries such as the United Kingdom, Australia, New Zealand, Japan and Ireland have historically maintained rabies-free status through strict import controls. Loss of that status would have major health and economic consequences.

3.2 The minimum post-vaccination period: immunological basis of the 21-day deadline

European Union legislation, as set out in Reglamento (UE) n.º 576/2013, establishes that rabies vaccination is not valid for international movement until 21 days have elapsed since administration of the first dose in a previously unvaccinated animal. This deadline is not arbitrary or merely bureaucratic. It has a direct basis in the kinetics of the humoral immune response.

Following administration of a rabies vaccine to an animal with no prior vaccination history, neutralising antibodies begin to be detectable approximately between days 7 and 14 post-vaccination in most animals, and reach levels considered protective in the range of 21 to 28 days. The 21-day deadline establishes the point from which the probability that an animal has generated an adequate immune response is sufficiently high for the purposes of population-level movement control.

It is important to understand that the 21-day period is a population control criterion, not an individual protection guarantee.

3.3 Booster vaccinations: the grace period and its implications

For animals with prior and valid rabies vaccination history, booster vaccination within the validity period of the previous vaccine does not restart the 21-day waiting period. The

anamnestic response is sufficiently rapid to consider the animal protected immediately after booster vaccination, which Reglamento (UE) 576/2013 explicitly recognises.

However, if booster vaccination is performed after the previous vaccine has expired, most international regulations treat that act as a first vaccination, with the consequent 21-day waiting period. Verification of this point is one of the most frequent sources of error in planning international travel with animals.

3.4 The RNATT: rabies antibody titration test

The Rabies Neutralising Antibody Titre Test (RNATT) is a serological test that quantifies the concentration of neutralising antibodies against rabies virus in blood serum. The international reference threshold, established by the OIE (World Organisation for Animal Health) and adopted by the main destinations that require it, is 0,5 UI/mL.

The RNATT does not measure immunity in a broad sense --it does not evaluate the cellular response, nor long-lasting immunological memory--. It specifically measures the concentration of circulating neutralising antibodies at the time of sample collection. The 0,5 UI/mL threshold was established as a reference cut-off value based on correlation studies between serological titres and protection in experimental challenge models.

Countries such as Australia, New Zealand, Japan, the United Kingdom and other rabies-free destinations require the RNATT as part of the import protocol. The sequence: the RNATT must be performed in an OIE-accredited laboratory, from a sample collected at least 30 days after valid rabies vaccination, and the result must meet the 0,5 UI/mL threshold. A result below the threshold requires revaccination and repetition of the test.

Key distinction: vaccination vs. serological test Rabies vaccination certificate: documents that the vaccination act was performed. Verifies the clinical event, not the immunological outcome. $RNATT \geq 0,5 \text{ UI/mL}$: documents that the animal's immune system responded to vaccination by generating neutralising antibodies at sufficient levels. Verifies the immunological outcome, not the clinical act. Both documents are required in destinations that require the RNATT. Neither replaces the other.

Section 4 Identification and correct sequence

4.1 The microchip as a documentary prerequisite

The relationship between microchip and vaccination in the context of international movement is not one of simple coexistence: it is one of sequential dependence. The microchip must be implanted and its number verified before the rabies vaccine intended to be recorded in the export documentation is administered. This sequence is not convention; it has its basis in the logic of traceability.

If the vaccine is administered before microchip implantation, a technically insoluble question arises: how can it be established that the animal carrying the subsequently implanted microchip is the same individual to whom the vaccine was previously administered? Most international regulations --including Reglamento (UE) n.º 576/2013 and the requirements of UK, Australia and New Zealand-- consider vaccination performed before microchip implantation invalid for import purposes.

4.2 Traceability: the number that links everything

The 15-digit ISO microchip number is the central identifier of the entire documentary chain of the animal in international transit. That number must appear exactly and identically in each document: vaccination certificate, RNATT results when applicable, health certificate or European passport, antiparasitic treatment certificates, and any other destination document.

A single discrepancy in the number may invalidate the entire documentary chain. Verification of the number by direct chip reading, compared with that recorded in the system and that printed on the documentation, is a control step that should not be omitted.

4.3 Documentary consequences of incorrect sequence

The sequence error --vaccination before microchip-- is one of the few errors in the export process that is technically irreversible in the short term. It cannot be corrected by a retroactive declaration or by a new vaccination simultaneous with chip implantation. The

animal must receive the microchip and start the vaccination protocol again, with the 21-day waiting period, and in destinations requiring RNATT complete the serological test. Depending on the destination, total delay may be several months.

Section 5 Biological vs. legal validity

5.1 Real immunological duration vs. duration recognised by regulation

The duration of immunity conferred by a vaccine is a biological variable. The duration of legal validity, in contrast, is a regulatory decision. These two durations do not always coincide. In the case of the rabies vaccine, modern vaccines have demonstrated immunity durations that may exceed three years in correctly vaccinated dogs. However, many jurisdictions maintain annual or biennial booster cycles for reasons that combine the precautionary principle and the social function of the annual veterinary visit.

In the context of international movement, the legal validity of the vaccination certificate always prevails over individual immunological assessment.

5.2 The difference between population management and individual immunity

Health control systems are designed to manage risk at the population level. The objective is not to guarantee that each individual animal is immune, but to keep the probability of pathogen introduction into a recipient population below an acceptable threshold. An animal that has passed all regulatory requirements is not, by definition, free of individual biological risk. It may have experienced undetected primary vaccine failure or be in the incubation period of a disease acquired just before travel.

5.3 Regulatory differences between destinations

The duration of validity recognised for the rabies vaccination certificate varies between jurisdictions. A certificate with recognised validity in the country of origin may have different validity --or have expired-- under the legislation of the country of destination. Validity must always be calculated according to destination criteria, not those of origin.

Section 6 Formats and regulatory framework by country

Regulatory update warning -- Section 6 The information in this section describes the type of regulatory framework of each jurisdiction, not the current specific requirements. Regulations change frequently. Official formats, exact deadlines, accredited laboratories and validation procedures may be modified without prior notice. Direct and up-to-date verification with the competent health authority of the country of destination and of transit is mandatory before issuing any certificate. This article does not replace that process.

6.1 European Union -- Supranational harmonisation framework

The European Union has the most harmonised companion animal movement system in the world. Reglamento (UE) n.º 576/2013 establishes a uniform framework applicable to all 27 member states. Reference document: the European Pet Passport, issued by authorised veterinarians. Includes identification (microchip), valid rabies vaccination, antiparasitic treatments when required. The minimum period post-first rabies vaccination is 21 days.

For movement from third countries to the EU: an official health certificate is required according to the models established in the applicable implementing regulations, issued by the official veterinarian of the country of origin. Verification of the current model is the responsibility of the issuing veterinarian.

6.2 United Kingdom -- Post-Brexit certification

The reference document for entry to the United Kingdom from abroad is the Animal Health Certificate (AHC), issued by an accredited official veterinarian a maximum of 10 days before travel. The AHC is not reusable: each trip requires a new certificate. ISO 11784/11785 microchip mandatory. Verification of the current model on GOV.UK is mandatory before issue.

6.3 Australia -- DAFF

Australia applies one of the strictest biosecurity protocols in the world. It requires rabies vaccination, $RNATT \geq 0,5$ UI/mL in an accredited laboratory, post-RNATT waiting period, specific antiparasitic treatments and mandatory quarantine on arrival. Sequence: microchip before rabies vaccination; vaccination before RNATT, with a minimum of 30 days between vaccination and sample collection for the RNATT. Verification on agriculture.gov.au before issue is mandatory.

6.4 New Zealand -- MPI New Zealand

New Zealand applies the CATDOG.GEN standard, one of the most demanding protocols worldwide. Sequence: microchip → rabies vaccination → RNATT → post-RNATT waiting period → treatments → certification → mandatory quarantine. Particularity: the microchip must be implanted before sample collection for the RNATT, not only before vaccination. Verification on mpi.govt.nz before issue is mandatory.

6.5 Other destinations -- Mandatory verification principle

For all destinations not detailed, verification with the competent health authority of the country of destination is mandatory. The universal operational principle:

The central message is that correct regulatory interpretation and verification of current official requirements at the time of issue are the only reasonable guarantee of documentary compliance.

Section 7 Technical errors that compromise certificate validity

7.1 Microchip-certificate inconsistency

The discrepancy between the microchip number read on the animal and that recorded in the documents is the error with the greatest impact. Frequent variants: number transcribed with typographical error; number recorded from the chip packaging without post-implantation verification; number of an old chip with a second chip implanted without document update; different number in different documents of the same dossier.

7.2 Dates incompatible with minimum periods

The most frequent incompatibilities: rabies vaccination date after the health certificate issue date; sample collection date for RNATT before 30 days post-vaccination; certificate issued outside the required time window; rabies vaccine date before microchip implantation.

7.3 Vaccines unauthorised in destination

Not all vaccines available in the country of origin are authorised or recognised in the country of destination. Verification that the vaccine administered is on the list of vaccines recognised by the destination is a prior step to choosing the vaccine.

7.4 Absence of signature, stamp or authorisation number

Many destinations require signature by a veterinarian with specific official authorisation, registration number, authorisation number and establishment stamp. Absence of any of these elements may be cause for formal rejection.

7.5 Incomplete recording of batch and manufacturer

Some destinations require recording of the batch number and manufacturer on the certificate. Omission when required is cause for invalidation. Recording at the time of vaccination --not afterwards-- guarantees its availability and accuracy.

Section 8 Evidence limitations

8.1 The certificate does not guarantee absolute individual immunity

The documentary validity of the certificate and individual immunological protection are two related but not equivalent variables. A formally correct certificate certifies that the required medical-documentary acts were performed. It does not certify that the immune system responded optimally or that the cold chain was perfect. This does not devalue it as an absolute individual guarantee, which was never its function.

8.2 Absence of studies on transport impact on vaccine response

There are no published studies that have evaluated whether the stress of air transport has a measurable impact on vaccine antibody titres in dogs or cats in the immediate post-travel period. In the absence of that evidence, it is not possible to state that transport compromises the vaccine response, nor that it does not.

8.3 Regulation as an instrument of collective risk management

Regulatory requirements are instruments of health policy, not individualised clinical protocols. The veterinarian working in travel medicine operates at the intersection between the individual biology of their patient and the collective regulatory framework.

Section 9 Final synthesis

The vaccination certificate for international movement is the administrative translation of a complex immunological event. Its validity results from the precise alignment of three dimensions that must be fulfilled simultaneously:

The clinical dimension: the vaccination act must have been performed correctly, with the appropriate vaccine, under the required technical conditions. The veterinarian's signature declares that this dimension was fulfilled.

The identity dimension: the animal must be unequivocally identified before vaccination by ISO microchip, and that identifier must be accurately recorded in all documents of the health dossier.

The regulatory dimension: the format, deadlines, required contents and authorised signatures must correspond to what the current regulations of the country of destination establish at the exact time of issue.

When these three dimensions are aligned, the certificate fulfils its function. When any fails --through clinical error, identification defect or regulatory deviation-- the document loses that certification capacity. This interdependence is what makes the international vaccination certificate a complex medical-documentary act.

References

- Unión Europea. Reglamento (UE) n.º 576/2013 del Parlamento Europeo y del Consejo, de 12 de junio de 2013. Diario Oficial de la Unión Europea, L 178, 1-26.
- United Kingdom Government. The Microchipping of Dogs (England) Regulations 2015. SI 2015/108.
- United Kingdom Government. The Microchipping of Cats and Dogs (England) Regulations 2023.
- Australian Government. DAFF. Importing cats and dogs.
<https://www.agriculture.gov.au/biosecurity-trade/cats-dogs>
- New Zealand Government. MPI. Importing dogs and cats (CATDOG.GEN).
<https://www.mpi.govt.nz/importing/animals/importing-dogs-and-cats/>
- GOV.UK -- DEFRA/APHA. Bringing your pet dog, cat or ferret to Great Britain.
<https://www.gov.uk/bring-pet-to-great-britain>
- WOAH/OIE. Terrestrial Animal Health Code. Chapter on Rabies. <https://www.woah.org/en/what-we-do/standards/codes-and-manuals/terrestrial-code-online-access/>
- WSAVA. Guidelines for the Vaccination of Dogs and Cats (2016). Journal of Small Animal Practice, 57(1), E1-E45.
- Day, M. J., et al. (2016). WSAVA Guidelines for the vaccination of dogs and cats. JSAP, 57(1), E1-E45. https://doi.org/10.1111/jsap.2_12431
- Schultz, R. D., et al. (2010). Age and long-term protective immunity in dogs and cats. J Comp Pathol, 142(Suppl 1), S102-S108. <https://doi.org/10.1016/j.jcpa.2009.10.009>
- Tizard, I. R. (2022). Veterinary Immunology (11th ed.). Elsevier.
- WHO. Rabies -- Key facts. <https://www.who.int/news-room/fact-sheets/detail/rabies>
- Rupprecht, C. E., et al. (2017). Rabies vaccines and post-exposure prophylaxis. In Plotkin's Vaccines (7th ed.). Elsevier.
- Cliquet, F., et al. (1998). Development of a fluorescent antibody virus neutralisation test (FAVN test). J Immunol Methods, 212(1), 79-87.
[https://doi.org/10.1016/S0022-1759\(97\)00212-3](https://doi.org/10.1016/S0022-1759(97)00212-3)

Figure 1. Decision flowchart — certificate validity for international pet movement.

Sequential AND-logic: every condition must hold; any single failure invalidates the whole documentary chain. Adapted from EU Reg. 576/2013

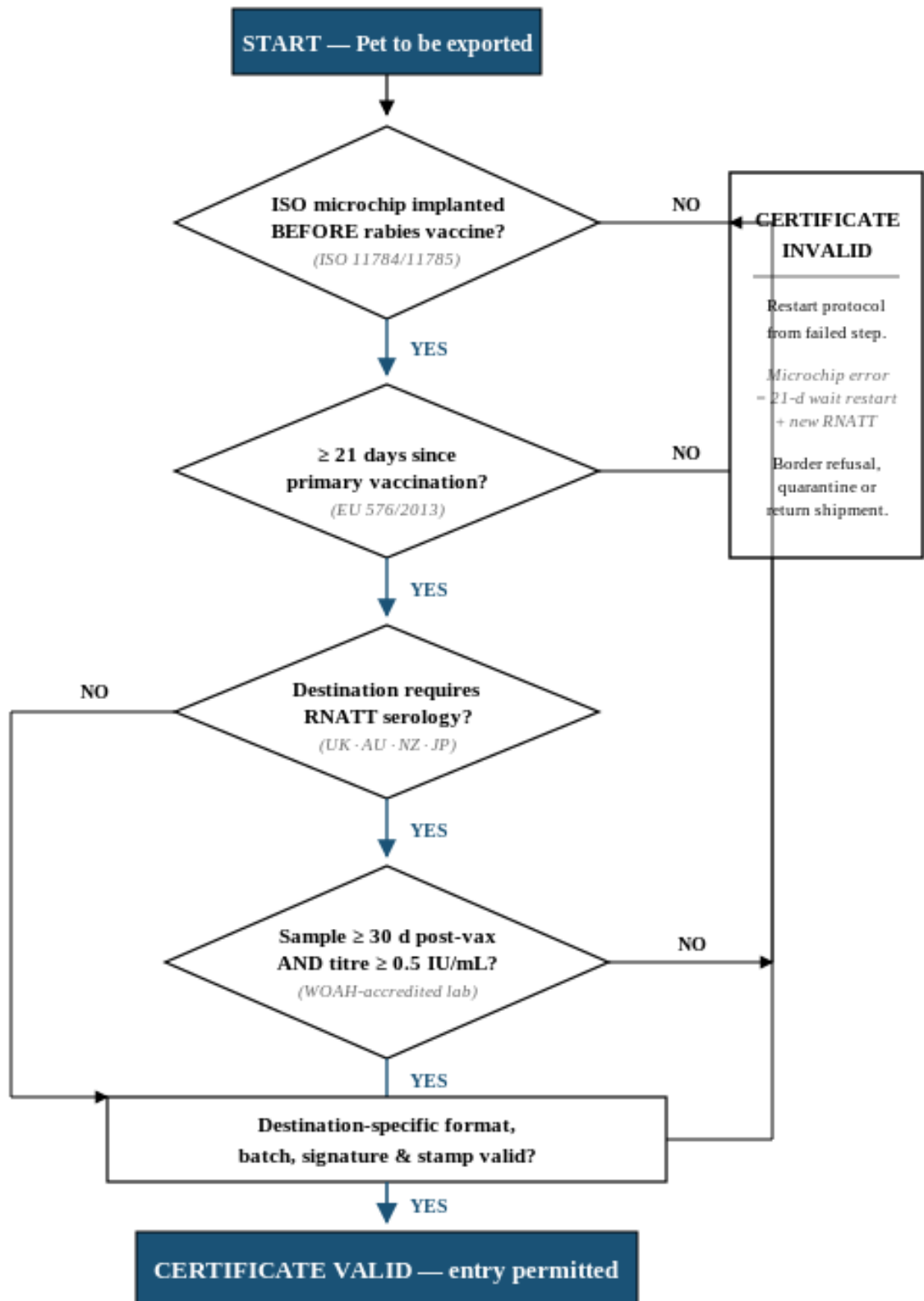


Figure 2. Comparative matrix — certificate validity windows by destination.

Synthesis of EU Reg. 576/2013, UK APHA AHC scheme, AU DAFF, NZ MPI CATDOG.GEN. Subject to regulatory updates — direct verification with competent authority is mandatory before emis.

Requirement	European Union <i>Reg. 576/2013</i>	United Kingdom <i>APHA — AHC</i>	Australia <i>DAFF</i>	New Zealand <i>MPI · CATDOG.GEN</i>	Other 3rd countries <i>verify per destination</i>
ISO microchip sequence <i>11784/11785</i>	Before vaccine <i>mandatory</i>	Before vaccine <i>mandatory</i>	Before vaccine <i>mandatory</i>	Before RNATT <i>stricter rule</i>	Per destination
Primary-vax wait <i>days post-injection</i>	≥ 21 d	≥ 21 d	≥ 21 d	≥ 21 d	Per destination
RNATT serology <i>≥ 0.5 IU/mL (FAVN/RFFIT)</i>	Non-listed 3rd countries only	Required for most 3rd countries	Required <i>sample ≥ 30 d post-vax</i>	Required <i>sample ≥ 30 d post-vax</i>	Variable
Post-titre waiting period <i>sample date → entry</i>	3 months <i>non-listed origin</i>	3 months <i>most 3rd countries</i>	≥ 180 d <i>after 2nd titre</i>	≥ 180 d <i>strictest globally</i>	Variable
Certificate emission window <i>issue → travel</i>	EU Pet Passport <i>reusable, lifetime</i>	AHC ≤ 10 d <i>single use only</i>	Import permit <i>per consignment</i>	Import permit <i>per consignment</i>	Per destination
Arrival quarantine <i>if compliant</i>	None <i>on full compliance</i>	None <i>on full compliance</i>	Mandatory <i>PAQS facility</i>	Mandatory <i>MPI facility</i>	Variable
Minimum total protocol <i>vaccine → arrival</i>	~ 4 months	~ 4 months	~ 7–9 months	~ 7–9 months	Variable

■ High-stringency value / strictest case

Acronyms: RNATT — rabies neutralising antibody titre test; FAVN — fluorescent antibody virus neutralisation; AHC

Note: validity windows must be calculated against destination criteria, never origin. Direct verification with the competent authority is mandatory before each emission.

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ISO 11784:1996. Radio Frequency Identification of Animals -- Code Structure.

ISO 11785:1996. Radio Frequency Identification of Animals -- Technical Concept.

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